



Unfiltered: Your Emotional Companion

Android Development Project Proposal

Team Members: **Kaushik Sandeep Joshi, Ashish Sandeep Joshi, Vidney Milind Jadhav.**

Course: CS 501 E1: Mobile Application Development

Project Overview – Unfiltered



What We Are Building

1

- A **mobile-first emotional support app** for Gen-Z
- Helps users **express, reflect, and manage emotions**
- Focus on **safe, anonymous, and non-judgmental experience**

2



Target Users

- Gen-Z (students & young professionals)
- Experiencing stress, anxiety, loneliness, burnout



Why It Matters

3

- Existing platforms increase comparison & stress
- Lack of **safe emotional spaces** for youth
- Provides **accessible daily emotional support**

Mobile-First Justification



Real-time emotional support



Uses sensors (mic, GPS, activity)



Habit-based daily usage



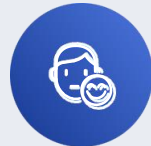
Private personal device

Emotional support requires instant, personal, context-aware access → mobile is essential.

Core Features



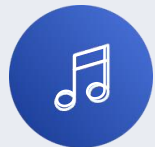
Journaling (text + mood tag)



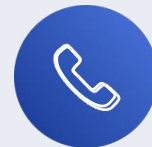
Daily mood tracking



Emotional rooms
(anonymous)



Music recommendation (planned)



Digital detox awareness

Core App Functionality & Compose UI

Multiple Screens (Compose)

- Login, Signup, Mood, Journal, Rooms
- Navigation using **NavHost**

UI Layouts

- **Scaffold** (structure + bottom bar)
- **LazyColumn** (journal list)
- Column / Box layouts

Material 3

- Buttons, Cards, NavigationBar
- Consistent **MaterialTheme**

Clean & Usable UI

- Simple user flow
- Interactive features (mood + journal)

State Management & Architecture

ViewModel-Based Design

- **Auth, Journal, Mood ViewModels**
- Handles logic outside UI

Reactive State

- **StateFlow** → **collectAsState()**
- Real-time UI updates

Clean Separation (MVVM)

- UI (Compose) ↔ ViewModel ↔ Repository

State Hoisting

- State lifted to ViewModel
- UI stays **simple & reusable**

Navigation & App Flow

NavHost-based navigation

Multiple connected screens: Login, Signup, Mood, Journal, Rooms, Detox

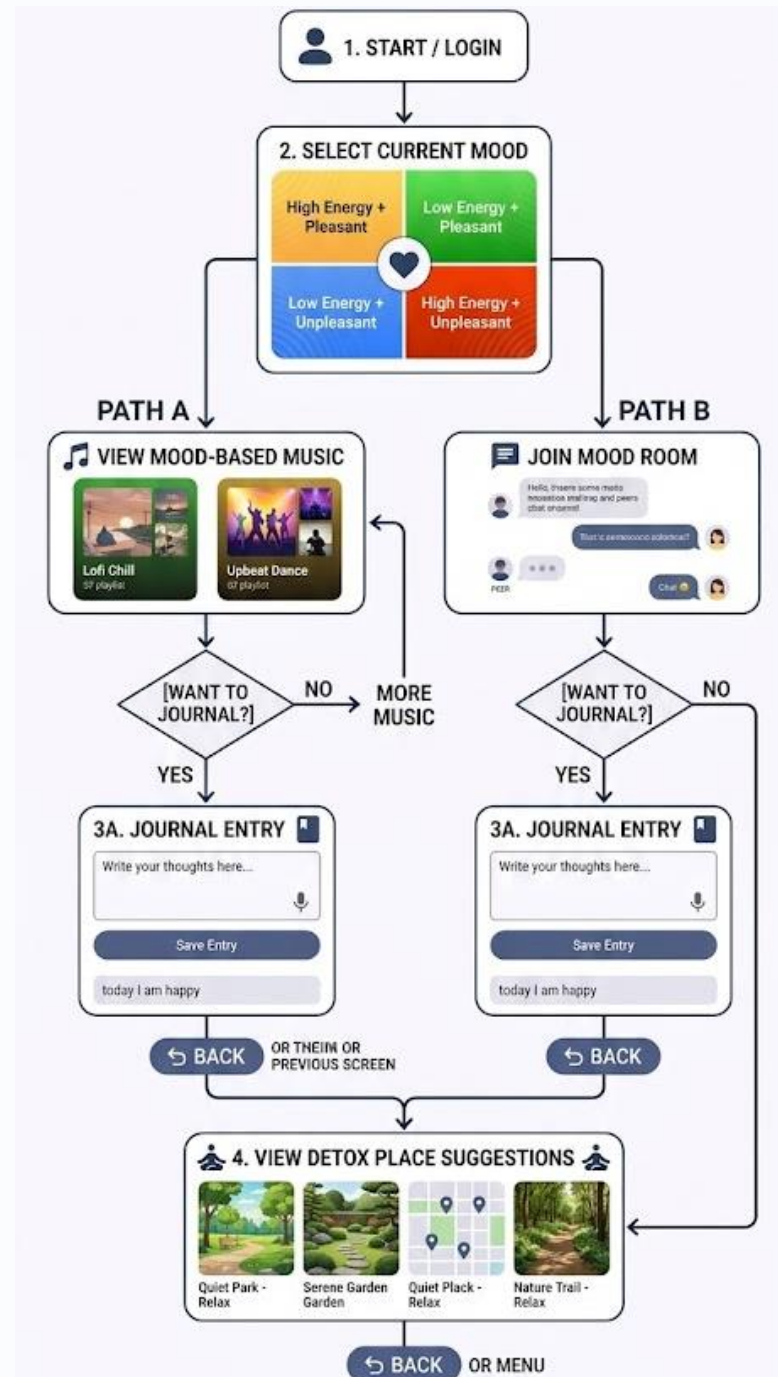
Clear user flow: **Login** → **Mood Selection** → **Feature Screens**

Argument passing with typed routes

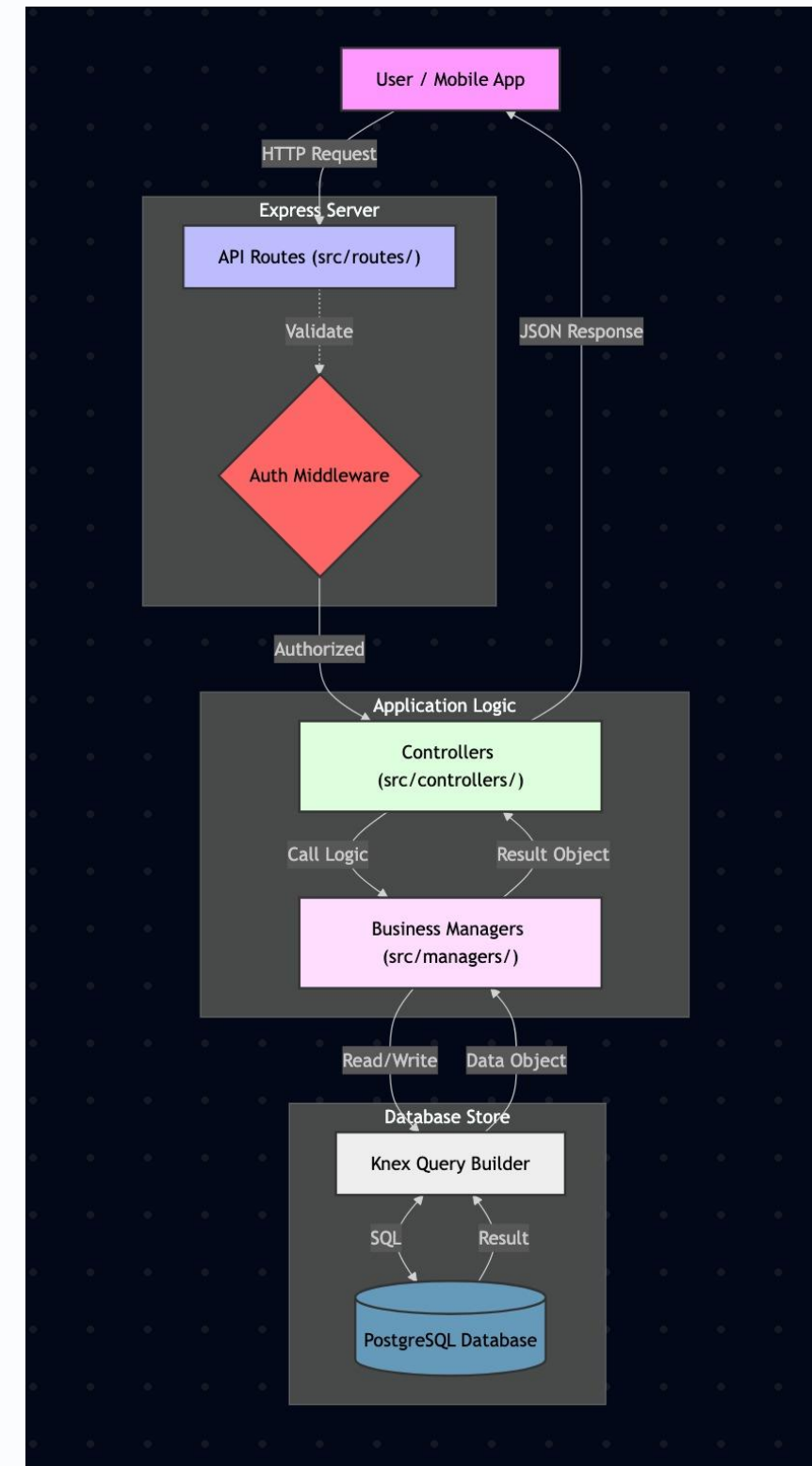
Example: **MoodCategory**
→
**MoodSubSelection(mod
eType)**

Bottom navigation for quick access to core features

App Flow



Architecture



Integration Progress Toward Requirements

API Integration

- Express backend with **Auth APIs (login/register)**
- Ready for app integration (Retrofit planned)

Data Persistence (*see diagrams*)

- **PostgreSQL + Knex schema implemented**
- Tables: users, journals, moods, rooms

Sensor Integration

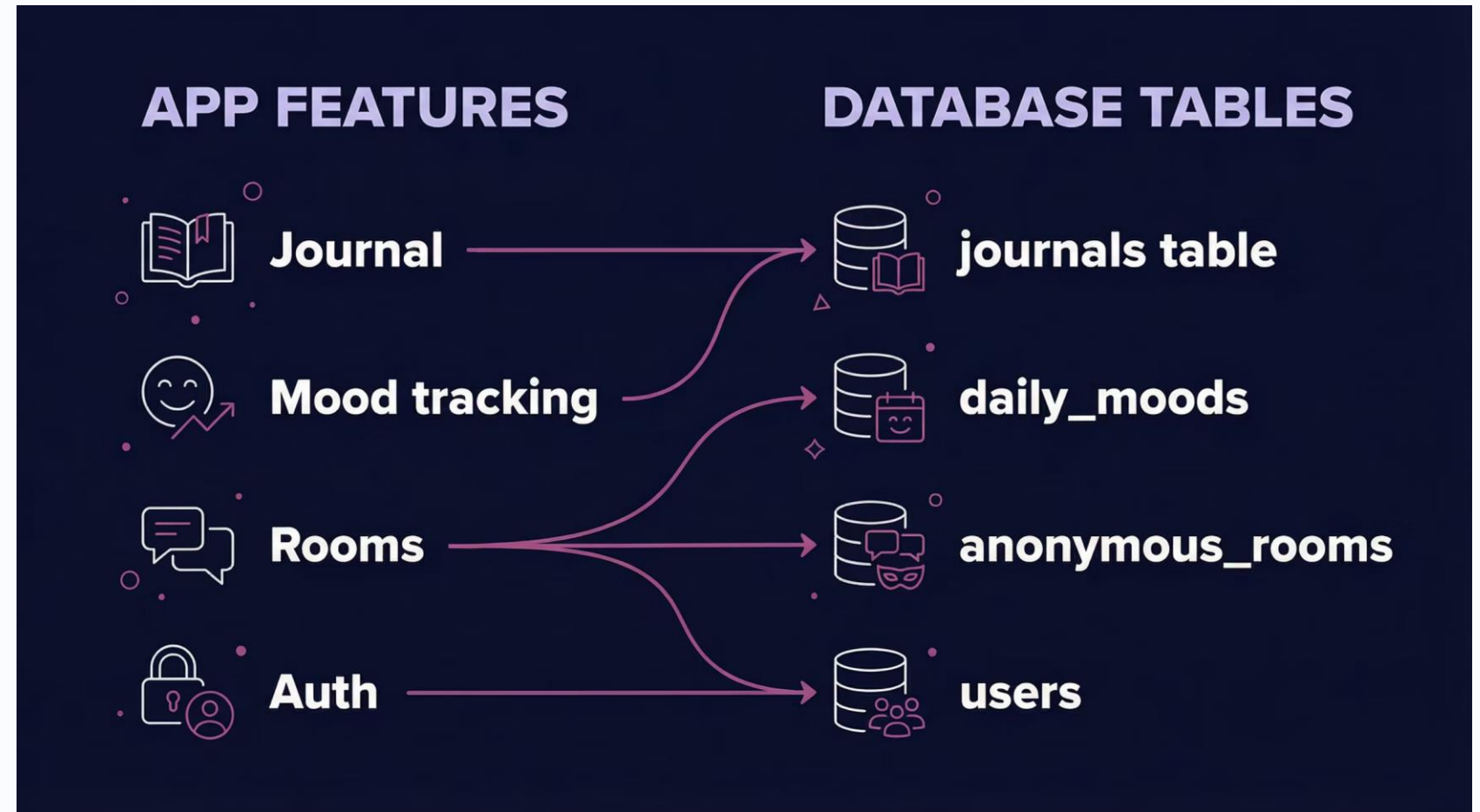
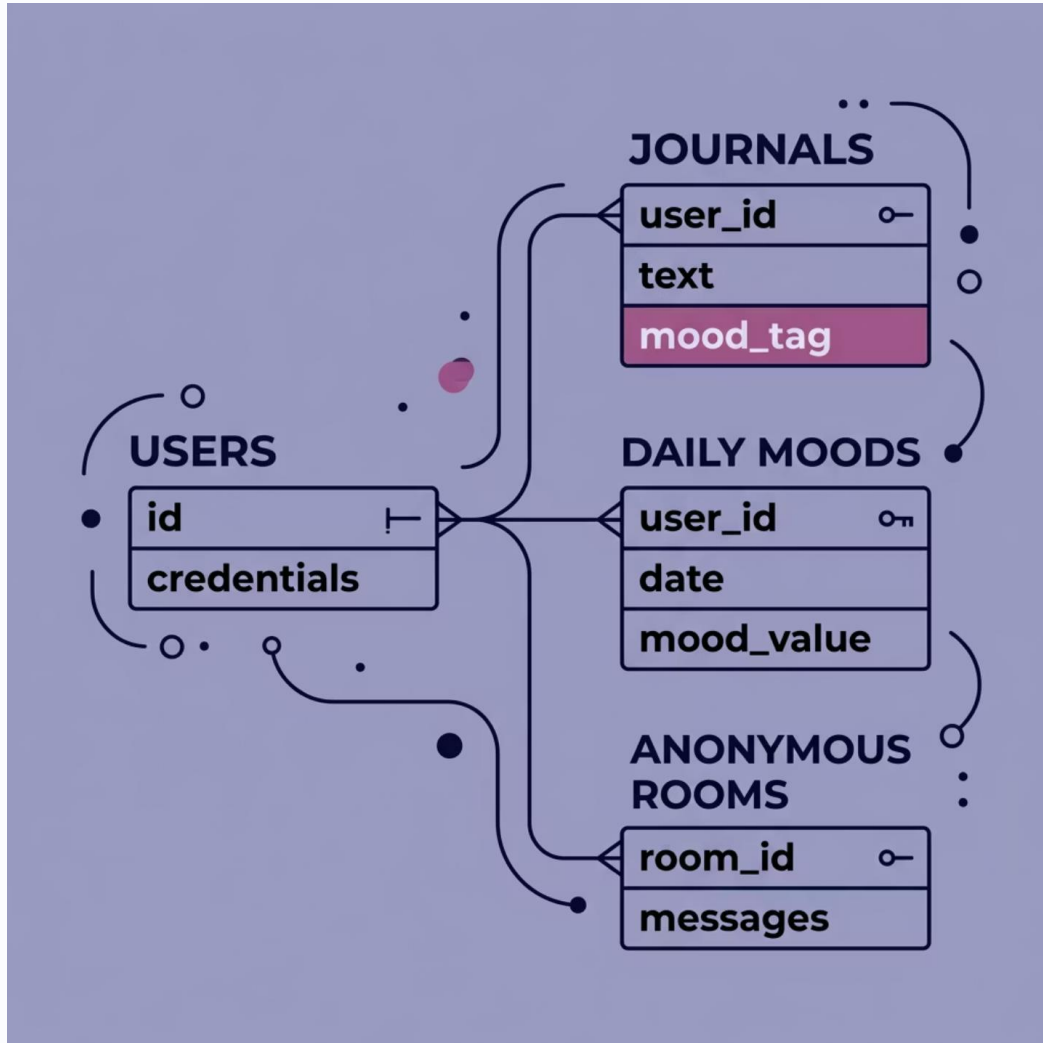
- **Microphone permission added**
- Voice journal feature in progress

Location / Maps

- **Google Maps API integrated**
- Detox location feature under development

Data Persistence

Backend schema using Knex



UI Implementation

Jetpack Compose UI

Scaffold layout

LazyColumn for lists

Material 3 components

Clean structured screens

UX Design



No likes / followers



No comparison



Calm UI design

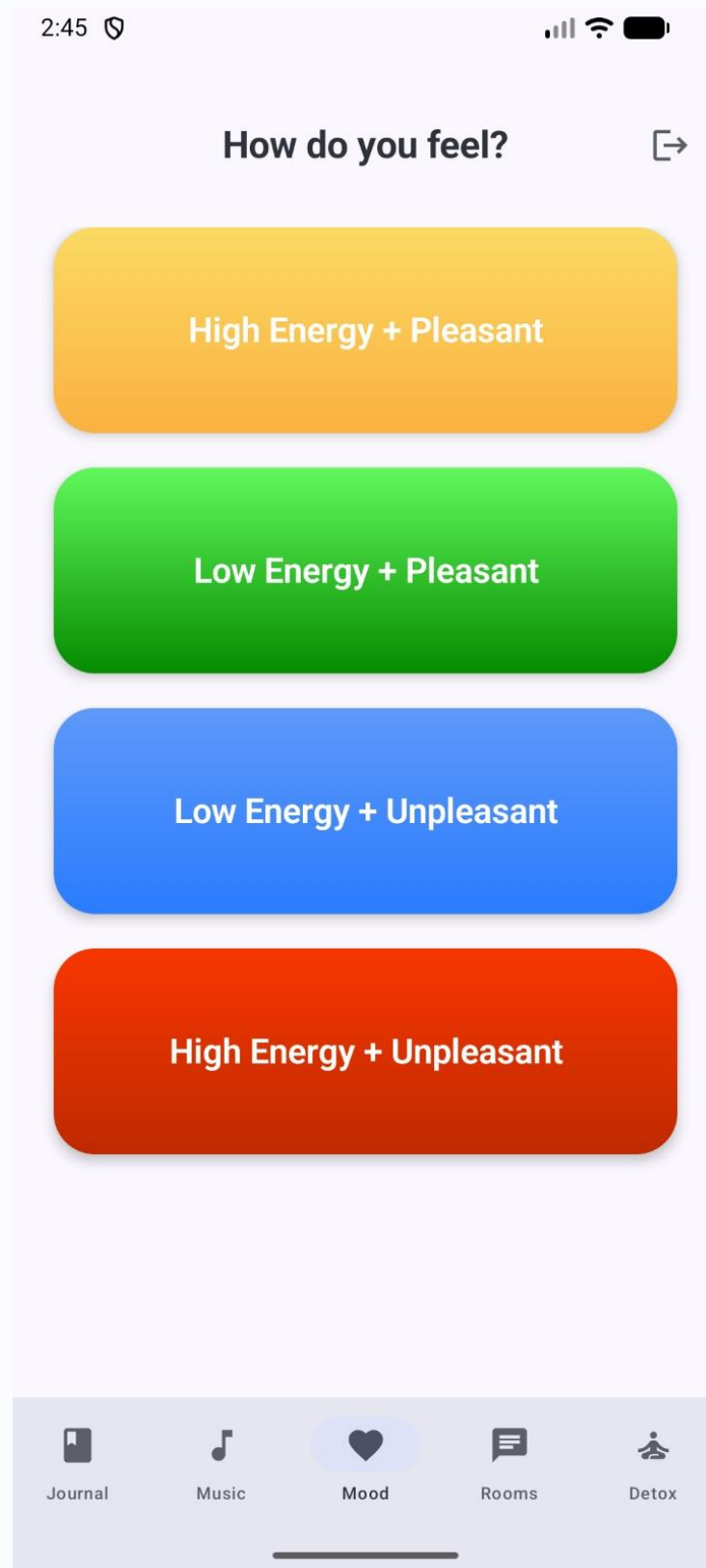


Short emotional interactions



Privacy-first

Designed for emotional safety, not engagement addiction.



UX Design

■ User-Centered Design

- Mood-based interaction (simple decision flow)
- Minimal steps to complete actions

■ Usability & Simplicity

- Clean layout with clear actions
- Large touch-friendly buttons

■ Responsive UI

- Built with **Jetpack Compose layouts**
- Adapts across screen sizes

■ Accessibility Considerations

- Color-coded mood selection
- Clear text and readable UI

Code Quality & Kotlin Usage



Clean & Modular Code

- Organized into **ViewModel**, **UI**, **Repository**
- Readable and maintainable structure



Kotlin Best Practices

- Idiomatic Kotlin (data classes, sealed classes)
- Null-safety and concise syntax



Coroutine Usage

- Used with **StateFlow**
- Handles async state updates efficiently

Responsible Use of AI

Practical Use in Development

- Debugging **Compose navigation & state issues**
- Understanding **StateFlow + ViewModel usage**
- Assisted with **code structure and best practices**

Critical Filtering

- Rejected **generic or irrelevant suggestions**

Controlled Usage

- AI used as a support tool, not a replacement



Team Members & Contributions



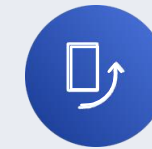
Kaushik

- Backend development (Auth APIs, database schema)
- API structure and integration setup



Ashish

- UI development using Jetpack Compose
- Screen design and user experience implementation



Vidney

- Navigation, state management, and ViewModel logic
- App flow and feature integration

Team Coordination

- Regular sync on features and flow
- Shared responsibilities across UI, logic, and backend
- Collaborative problem-solving

Project Progress

Done

- Core UI screens (Mood, Journal, Navigation)
- Navigation using **NavHost**
- ViewModel + **StateFlow integration**
- Journal feature (text)
- Mic sensor implemented
- **Database schema implemented** (users, journals, moods, rooms)
- **Backend Auth APIs** (login/register)
- **Google Maps API key configured**

Planned

- Mood analysis section
- Mood-based room filtering
- Spotify API integration (music recommendations)
- UI refinement & polish
- Maps feature (detox location logic)
- Voice journaling (mic functionality)

